

Solution Code - Engineering concepts - Algorithms_DataStructures - E_commerce_Platform_Search

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: SearchFunction.java

```
1 | import java.util.Arrays;
2 |
3 | class Product implements Comparable<Product> {
4 |     String id;
5 |     String name;
6 |
7 |     public Product(String id, String name) { this.id = id; this.name = name; }
8 |     public int compareTo(Product o) { return this.id.compareTo(o.id); }
9 | }
10 |
11 | public class SearchFunction {
12 |     public static int linearSearch(Product[] arr, String targetId) {
13 |         for (int i = 0; i < arr.length; i++) {
14 |             if (arr[i].id.equals(targetId)) return i;
15 |         }
16 |         return -1;
17 |     }
18 |
19 |     public static int binarySearch(Product[] arr, String targetId) {
20 |         int l = 0, r = arr.length - 1;
21 |         while (l <= r) {
22 |             int m = l + (r - l) / 2;
23 |             int cmp = arr[m].id.compareTo(targetId);
24 |             if (cmp == 0) return m;
25 |             if (cmp < 0) l = m + 1;
26 |             else r = m - 1;
27 |         }
28 |         return -1;
29 |     }
30 |     // Comparison: Linear Search is O(n), Binary Search is O(log n) but requires a sorted array.
31 | }
```